Lab Assignment no. 2

PRACTICAL NO. 5

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DIVISION :ET21

SUBJECT: EDS LAB

5.1.1.STACKED PLOT

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- **Create a stacked area plot using the data.**
- **Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.**

- **Display the plot showing the temperature variation for each city throughout the months of the year.**

CODE: import matplotlib.pyplot as

plt import pandas as pd

Data for Months and Temperature for three cities data

= {

'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December'],

'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],

'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],

'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]

}

Write your code... df=pd.DataFrame(data)

plt.stackplot(df['Month'],df['City_A_Tempe

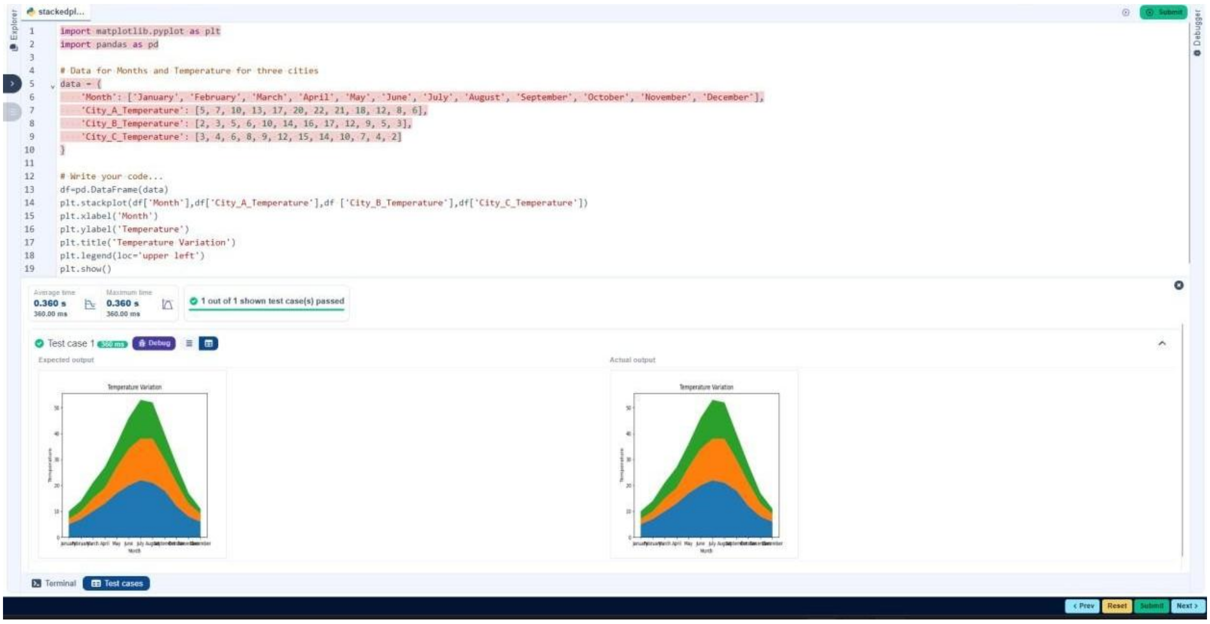
rature'],df

['City_B_Temperature'],df['City_C_Tempera
ture']) plt.xlabel('Month')

plt.ylabel('Temperature') plt.title('Temperature

Variation') plt.legend(loc='upper left')

plt.show()



5.1.2.TITANIC DATASET

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named titanic.csv and has been loaded using the pandas library. It contains the following columns:

- **Pclass: Passenger class (1 = First, 2 = Second, 3 = Third).**
- **Gender: Gender of the passenger (male/female).**
- **Age: Age of the passenger.**
- **Survived: Survival status (0 = Did not survive, 1 = Survived).**
- **Fare: Ticket fare paid by the passenger.**

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- **Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).**
- **Use the color skyblue for the bars.**
- **Title the plot as "Passenger Class Distribution".**
- **Label the x-axis as "Pclass" and the y-axis as "Count".**

Pie Chart (Gender Distribution):

- **Create a pie chart to display the distribution of male and female passengers.**
- **Use lightblue for males and lightcoral for females.**
- **Include percentages on the slices (use autopct='%1.1f%%').**
- **Title the plot as "Gender Distribution".**

Histogram (Age Distribution):

- **Create a histogram to visualize the distribution of passengers' ages.**
- **Use lightgreen for the bars with black edges (edgecolor = 'black').**
- **Set the number of bins to 8 for the histogram.**
- **Title the plot as "Age Distribution".**
- **Label the x-axis as "Age" and the y-axis as "Frequency".**

Bar Plot (Survival Count):

- **Create a bar plot to show the count of passengers who survived and those who did not, based on the Survived column.**
- **Use the colors lightblue for survivors (1) and lightcoral for non-survivors (0).**
- **Title the plot as "Survival Count".**
- **Label the x-axis as "Survived (0 = No, 1 = Yes)" and the y-axis as "Count".**

Scatter Plot (Fare vs Age):

- **Create a scatter plot to visualize the relationship between the Fare and Age of passengers.**
- **Use orange for the data points.**
- **Title the plot as "Fare vs Age".**
- **Label the x-axis as "Age" and the y-axis as "Fare".**

Code: import pandas as pd import

matplotlib.pyplot as plt

Load the Titanic dataset from the CSV file df

= pd.read_csv('titanic.csv')

Set up the figure for 5 subplots fig, axes = plt.subplots(3,

2, figsize=(12, 12))

write the code..

axes[0,0].bar(df['Pclass'].value_counts().index,df['Pclass'].value_counts(), color='skyblue') axes[0,

0].set_title("Passenger Class Distribution") axes[0, 0].set_xlabel("Pclass") axes[0,

0].set_ylabel("Count")

axes[0,1].pie(df['Gender'].value_counts(),labels=df['Gender'].value_counts().index,autop

ct='

%1.1f%%',colors=['lightblue','lightcoral']) axes[0,

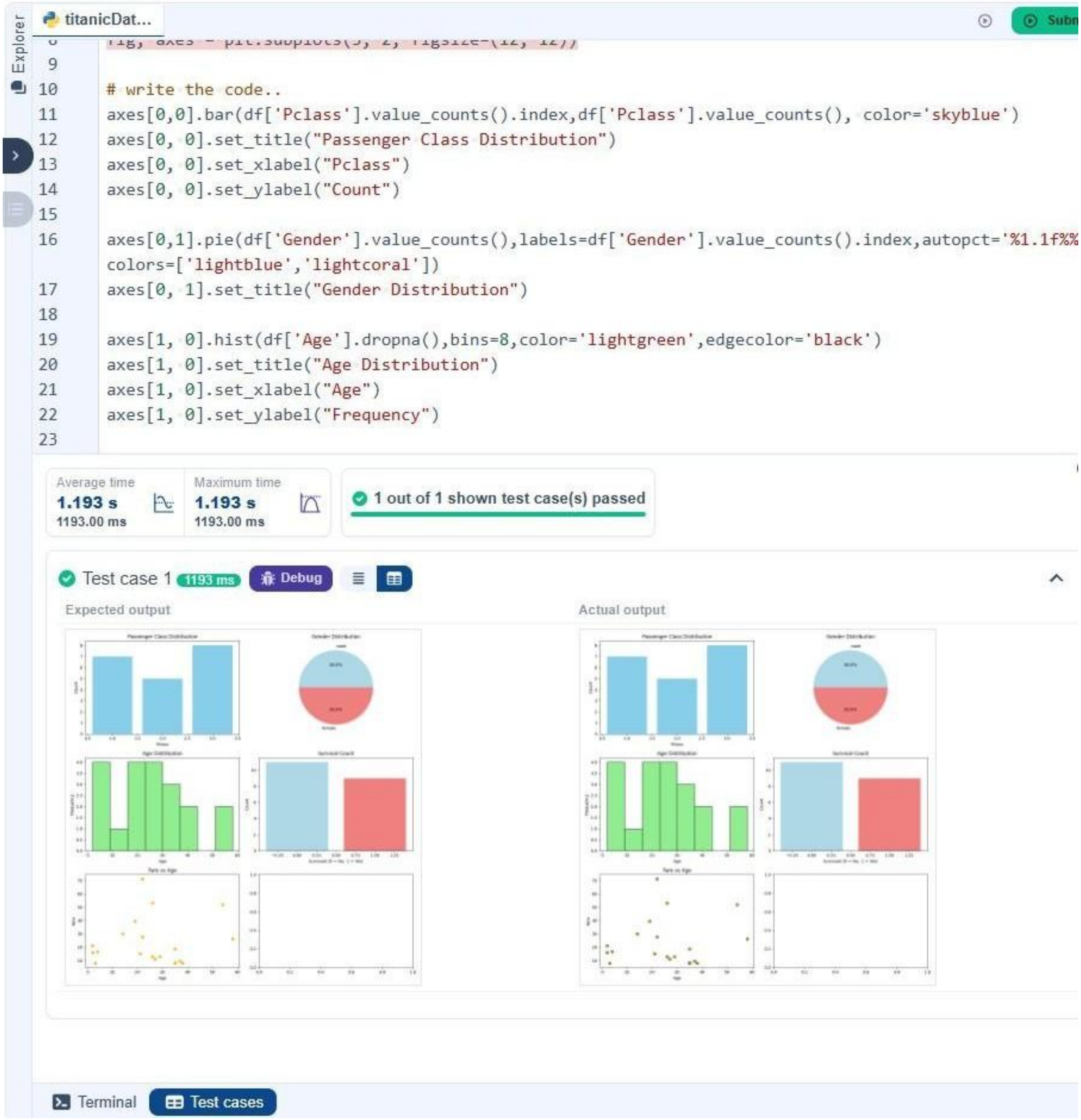
1].set_title("Gender Distribution")

axes[1, 0].hist(df['Age'].dropna(),bins=8,color='lightgreen',edgecolor='black')

axes[1, 0].set_title("Age Distribution") axes[1, 0].set_xlabel("Age") axes[1,

0].set_ylabel("Frequency")

```
axes[1,
1].bar(df['Survived'].value_counts().index,df['Survived'].value_counts(),color=['lightblue','ligh
tcoral']) axes[1, 1].set_title("Survival Count") axes[1, 1].set_xlabel("Survived (0 = No, 1 = Yes)")
axes[1, 1].set_ylabel("Count")

axes[2, 0].scatter(df['Age'], df['Fare'], color='orange', edgecolors='black')
axes[2, 0].set_title("Fare vs Age") axes[2, 0].set_xlabel("Age") axes[2,
0].set_ylabel("Fare")

plt.tight_layout() plt.show()
```



5.2.2.HISTOGRAM OF PASSENGER INFORMATION OF TITANIC

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. **Use 30 bins for the histogram.**
2. **Set the edge color of the bars to black (k).**
3. **Label the x-axis as 'Age' and the y-axis as 'Frequency'.**
4. **Add the title "Age Distribution" to the histogram.**

The Titanic dataset contains columns as shown below,

PassengerId	Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

import pandas as pd import matplotlib.pyplot

as plt

Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

Data Cleaning data['Age'].fillna(data['Age'].median(),

inplace=True) data['Embarked'].fillna(data['Embarked'

].mode()[0], inplace=True) data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':

0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],

drop_first=True)

Write your code here for Histogram plt.hist(data['Age'], bins=30,

edgecolor='k')

plt.title('Age Distribution')

plt.xlabel('Age')

plt.ylabel('Frequency')

plt.show()

Explorer

Histogra...

Subm

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Histogram
plt.hist(data['Age'], bins=30, edgecolor='k')

```

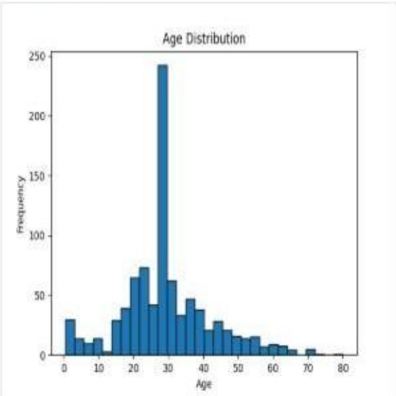
Average time **0.369 s** 369.00 ms

Maximum time **0.369 s** 369.00 ms

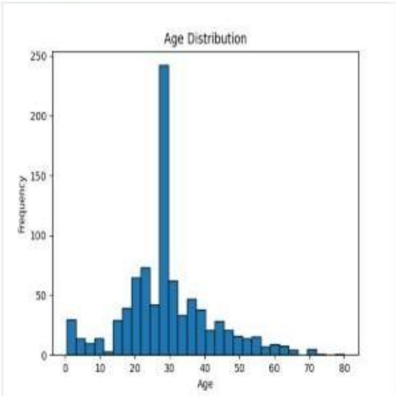
1 out of 1 shown test case(s) passed

Test case 1 369 ms Debug

Expected output



Actual output



Terminal Test cases

5.2.3.BAR PLOT OF SURVIVAL RATE OF PASSENGER

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. **Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).**
2. **Set the chart type to 'bar'.**
3. **Add the title "Survival Count" to the chart.**
4. **Label the x-axis as 'Survived' and the y-axis as 'Count'.**

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

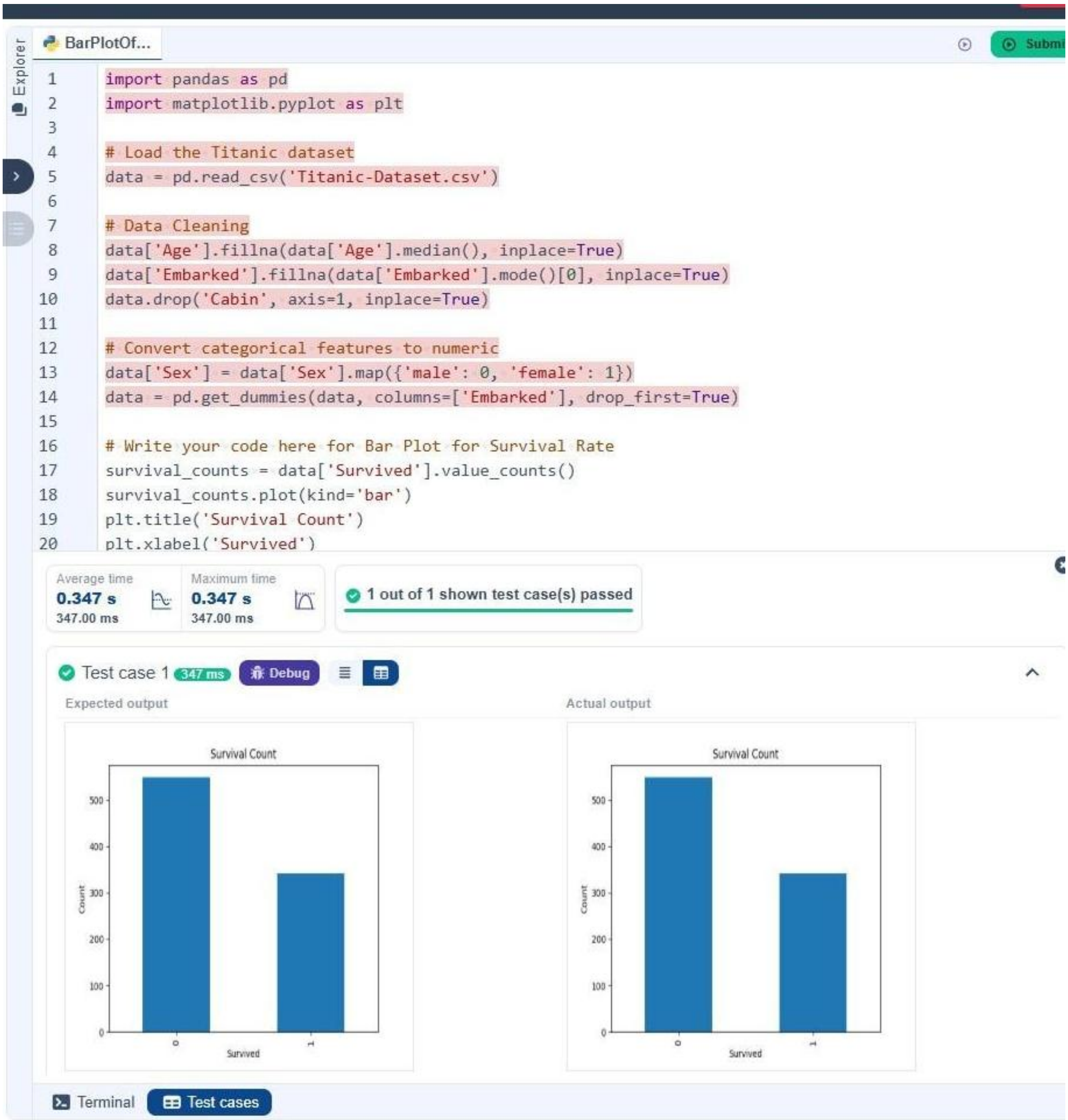
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot
as plt

# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':
0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# Write your code here for Bar Plot for Survival Rate
survival_counts = data['Survived'].value_counts()
survival_counts.plot(kind='bar') plt.title('Survival Count')
plt.xlabel('Survived') plt.ylabel('Count') plt.show()
```



5.2.4.BAR PLOT FOR SURVIVAL BY GENDER

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. **Group the data by the 'Sex' column, then use the value_counts() function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.**
2. **Use a stacked bar chart to display the survival counts.**
3. **Add the title "Survival by Gender" to the chart.**

4. **Label the x-axis as 'Gender' and the y-axis as 'Count'.**
5. **The legend should indicate 'Not Survived' and 'Survived'.**

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

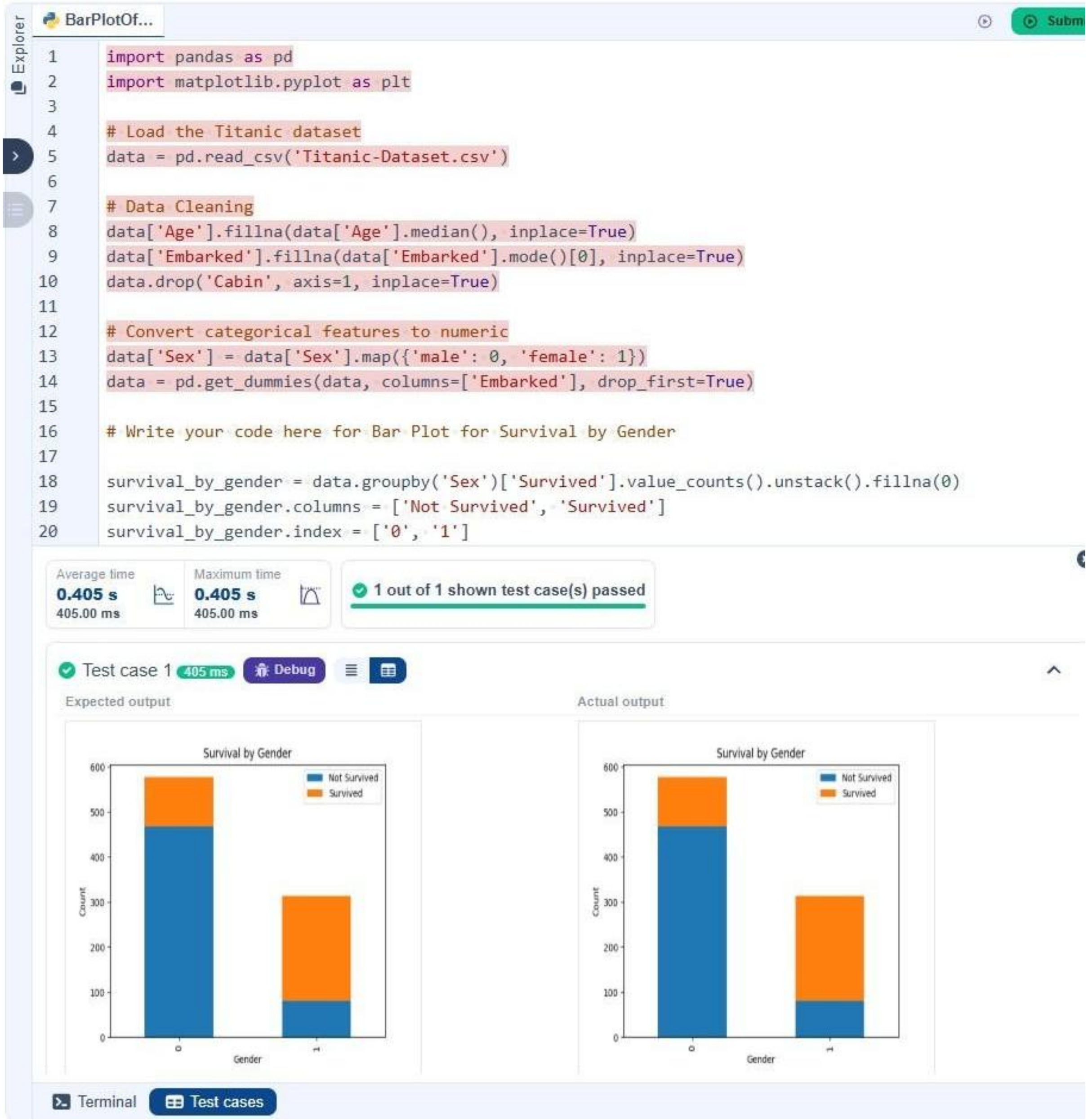
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot
as plt
```

```
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)


# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':

0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],

drop_first=True)


# Write your code here for Bar Plot for Survival by Gender


survival_by_gender = data.groupby('Sex')['Survived'].value_counts().unstack().fillna(0)

survival_by_gender.columns = ['Not Survived', 'Survived'] survival_by_gender.index = ['0',

'1'] survival_by_gender.plot(kind='bar', stacked=True) plt.title('Survival by

Gender') plt.xlabel('Gender') plt.ylabel('Count') plt.legend(title=None) plt.show()
```



5.2.5.BAR PLOT FOR SURVIVAL BY PCLASS

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. **Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().**
2. **Use a stacked bar chart to display the survival counts.**
3. **Add the title "Survival by Pclass" to the chart.**
4. **Label the x-axis as 'Pclass' and the y-axis as 'Count'.**

5. **The legend should indicate 'Not Survived' and 'Survived'.**

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

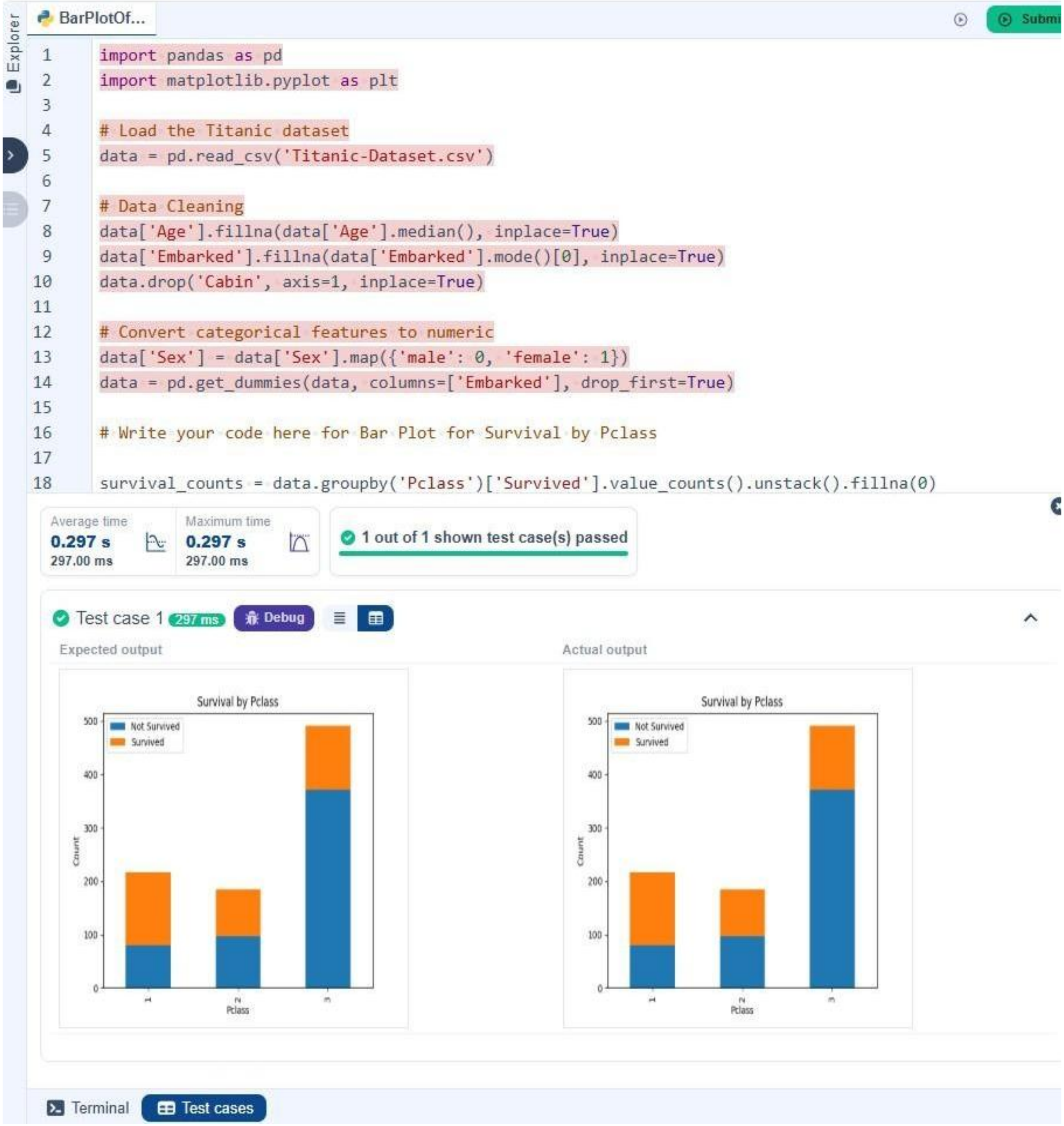
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot
as plt


# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':
0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# Write your code here for Bar Plot for Survival by Pclass

survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind='bar',stacked=True)

plt.title('Survival by Pclass') plt.xlabel('Pclass')
plt.ylabel('Count')

plt.legend(['Not Survived','Survived']) plt.show()
```



```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Pclass

survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)
```


5.2.6.BAR PLOT FOR SURVIVAL BY EMBARKED

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. **Use the Embarked column to determine the embarkation location. After converting this column into dummy variables (using pd.get_dummies()), plot the survival**

count based on the Embarked_Q column (representing passengers who embarked from Queenstown) in relation to survival.

2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked " to the chart.
4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

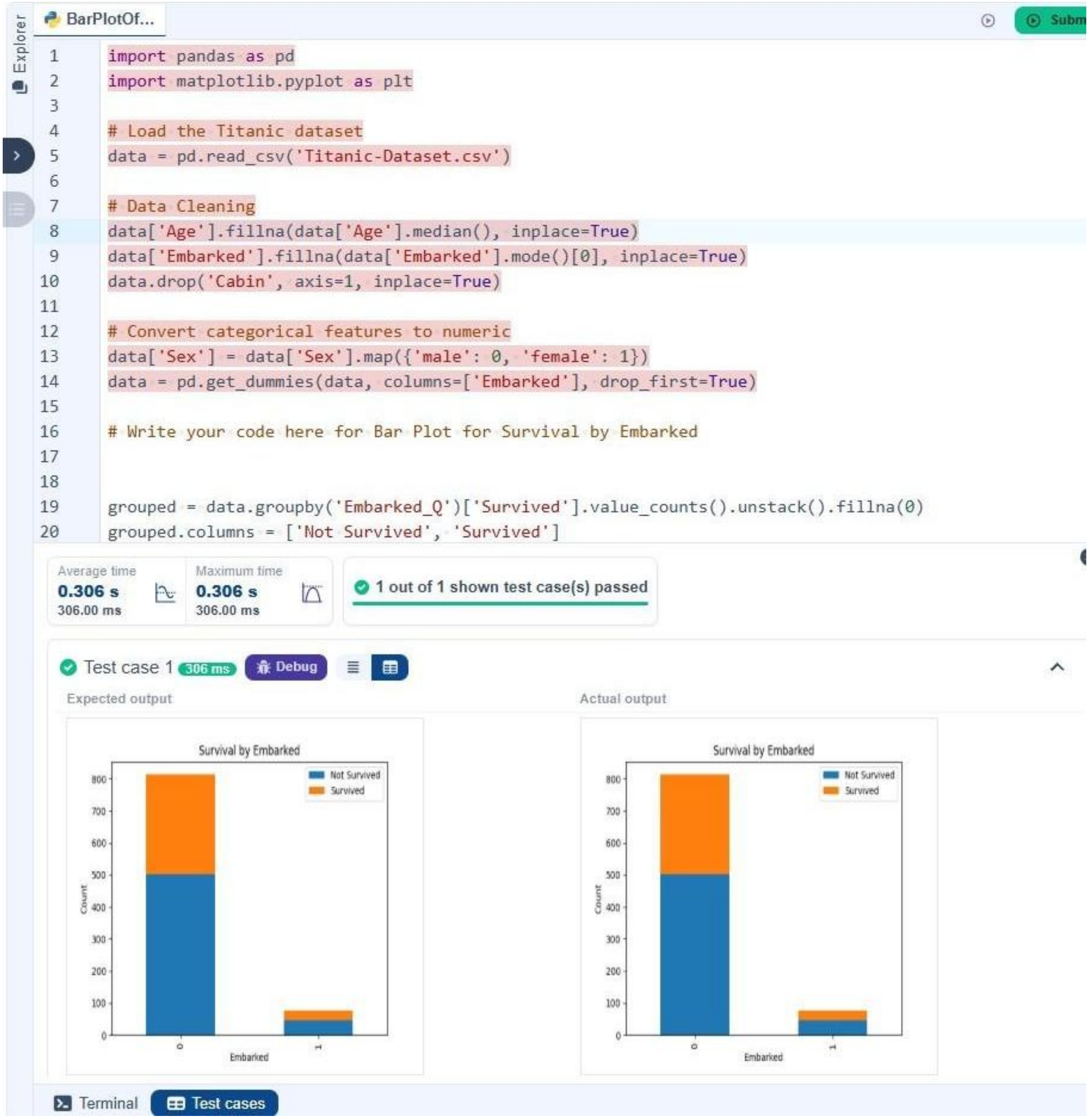
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd  import
matplotlib.pyplot as plt

# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':
0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# Write your code here for Bar Plot for Survival by Embarked

grouped = data.groupby('Embarked_Q')['Survived'].value_counts().unstack().fillna(0)
grouped.columns = ['Not Survived', 'Survived'] grouped.plot(kind='bar', stacked=True)
plt.title('Survival by Embarked') plt.xlabel('Embarked') plt.ylabel('Count')
plt.legend(title=None) plt.show()
```



5.2.7.BOX PLOT FOR AGE DISTRIBUTION

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. **Use the Pclass column to group the data for the boxplot.**
2. **Set the title of the plot to "Age by Pclass".**
3. **Remove the default subtitle with plt.suptitle('').**
4. **Label the x-axis as 'Pclass' and the y-axis as 'Age'.**

The Titanic dataset contains columns as shown below,

PassengerId	Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import

matplotlib.pyplot as plt


# Load the Titanic dataset data = pd.read_csv('Titanic-

Dataset.csv') # Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'
].mode()[0], inplace=True) data.drop('Cabin',
axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass



plt.figure(figsize=(8, 6)) data.boxplot(column='Age',
by='Pclass')  plt.suptitle('') plt.title('Age by Pclass')
plt.xlabel('Pclass') plt.ylabel('Age') plt.show()
```

Explorer

BoxPlotF...

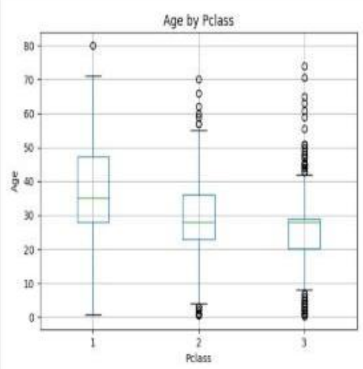
```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass

```

Average time 0.298 s 298.00 ms | Maximum time 0.298 s 298.00 ms

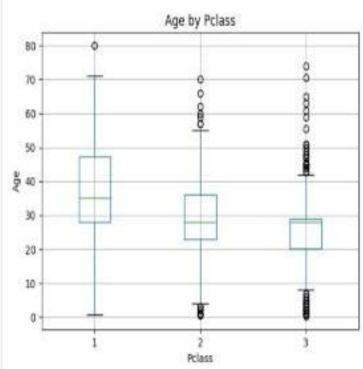
1 out of 1 shown test case(s) passed

Test case 1 298 ms Debug

Expected output



Actual output



Terminal Test cases

5.2.8.BOX PLOT FOR AGE BY SURVIVED

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. **Use the Survived column to group the data for the boxplot (0 = Did not survive, 1 = Survived).**
2. **Set the title of the plot to "Age by Survival".**
3. **Remove the default subtitle with plt.suptitle('').**
4. **Label the x-axis as 'Survived' and the y-axis as 'Age'.**

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

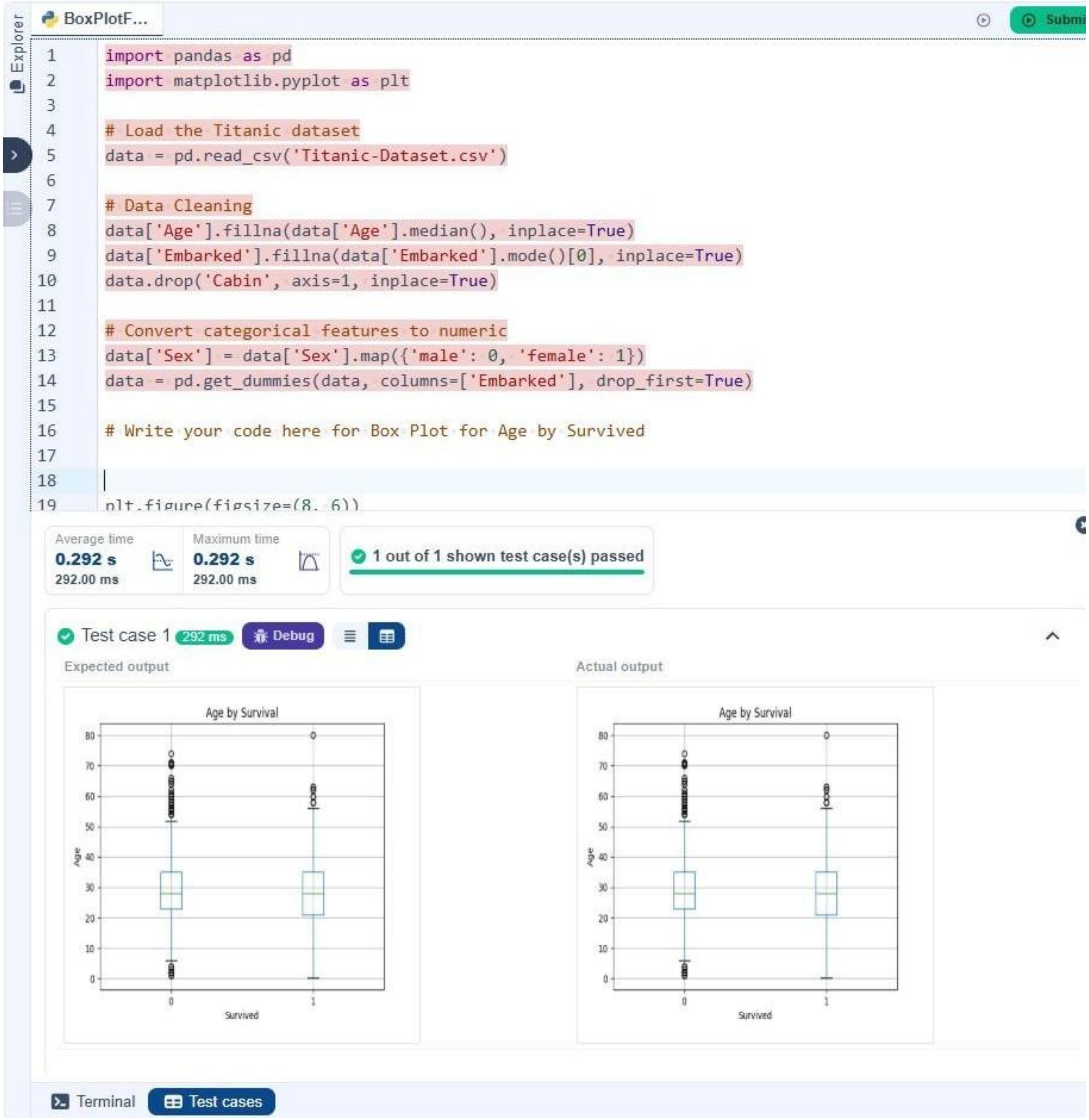
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot
as plt

# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':
0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# Write your code here for Box Plot for Age by Survived


plt.figure(figsize=(8, 6)) data.boxplot(column='Age',
by='Survived')  plt.suptitle('') plt.title('Age by
Survival') plt.xlabel('Survived') plt.ylabel('Age')
plt.show()
```



5.2.9. BOX PLOT FOR FARE BY PCLASSS

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. **Use the Pclass column to group the data for the boxplot.**
2. **Set the title of the plot to "Fare by Pclass".**
3. **Remove the default subtitle with plt.suptitle('').**
4. **Label the x-axis as 'Pclass' and the y-axis as 'Fare'.**

The Titanic dataset contains columns as shown below,

PassengerId	**Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked**

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

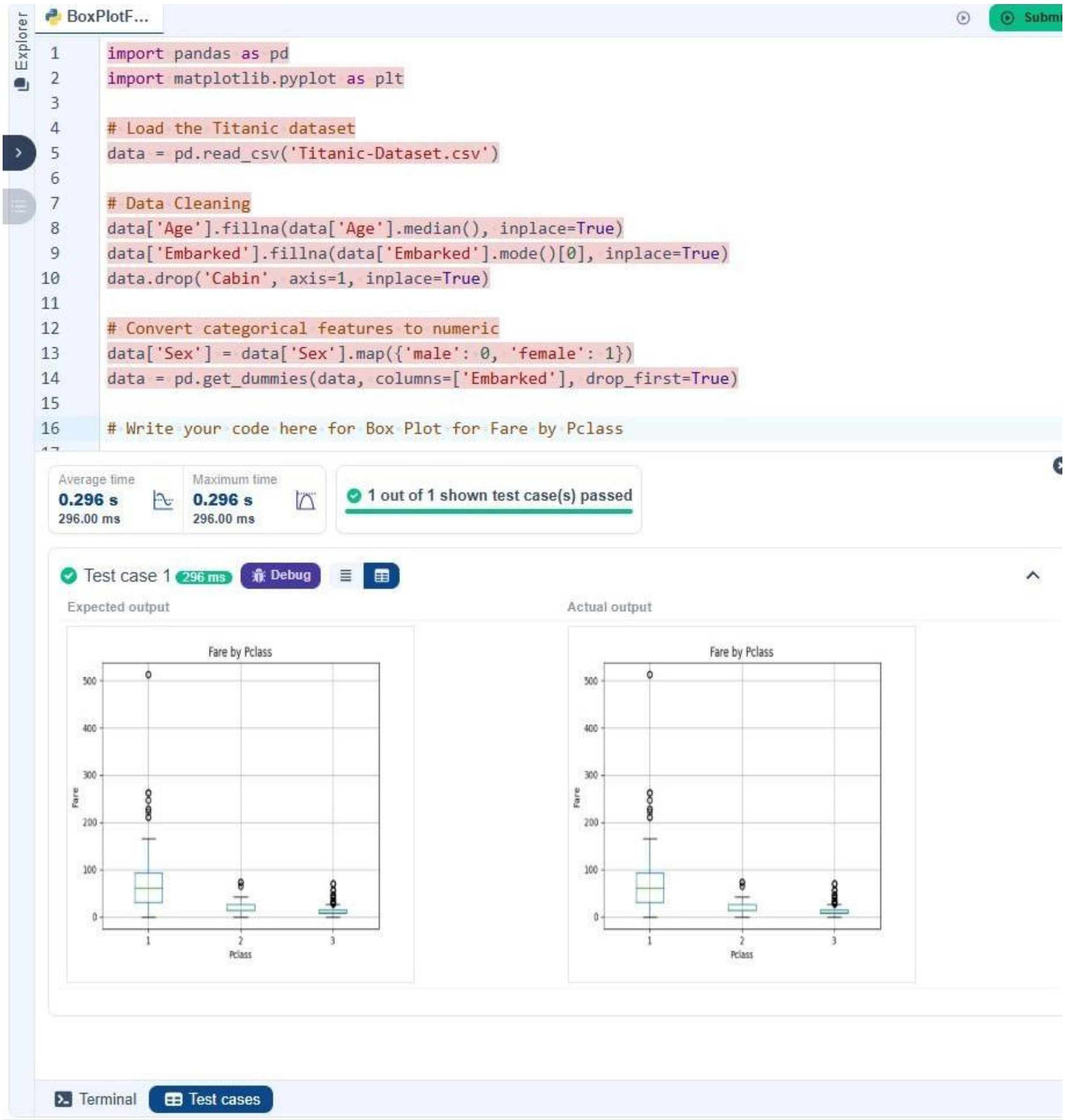
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot
as plt

# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

#
```

Data Cleaning

```
data['Age'].fillna(data['Age'].median(),

inplace=True)

data['Embarked'].fillna(data['Embarked'

].mode()[0], inplace=True) data.drop('Cabin',

axis=1, inplace=True)


# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':

0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],

drop_first=True)


# Write your code here for Box Plot for Fare by Pclass



plt.figure(figsize=(8, 6))

data.boxplot(column='Fare', by='Pclass')

plt.suptitle('') plt.title('Fare by Pclass')

plt.xlabel('Pclass') plt.ylabel('Fare') plt.show()
```



5.2.10.SCATTER PLOT FOR AGE VS. FARE

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. **Use the Age column for the x-axis and the Fare column for the y-axis.**
2. **Set the title of the plot to "Age vs. Fare".**
3. **Label the x-axis as 'Age' and the y-axis as 'Fare'.**

The Titanic dataset contains columns as shown below,

PassengerId	**Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked**

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot

as plt


# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

#
```

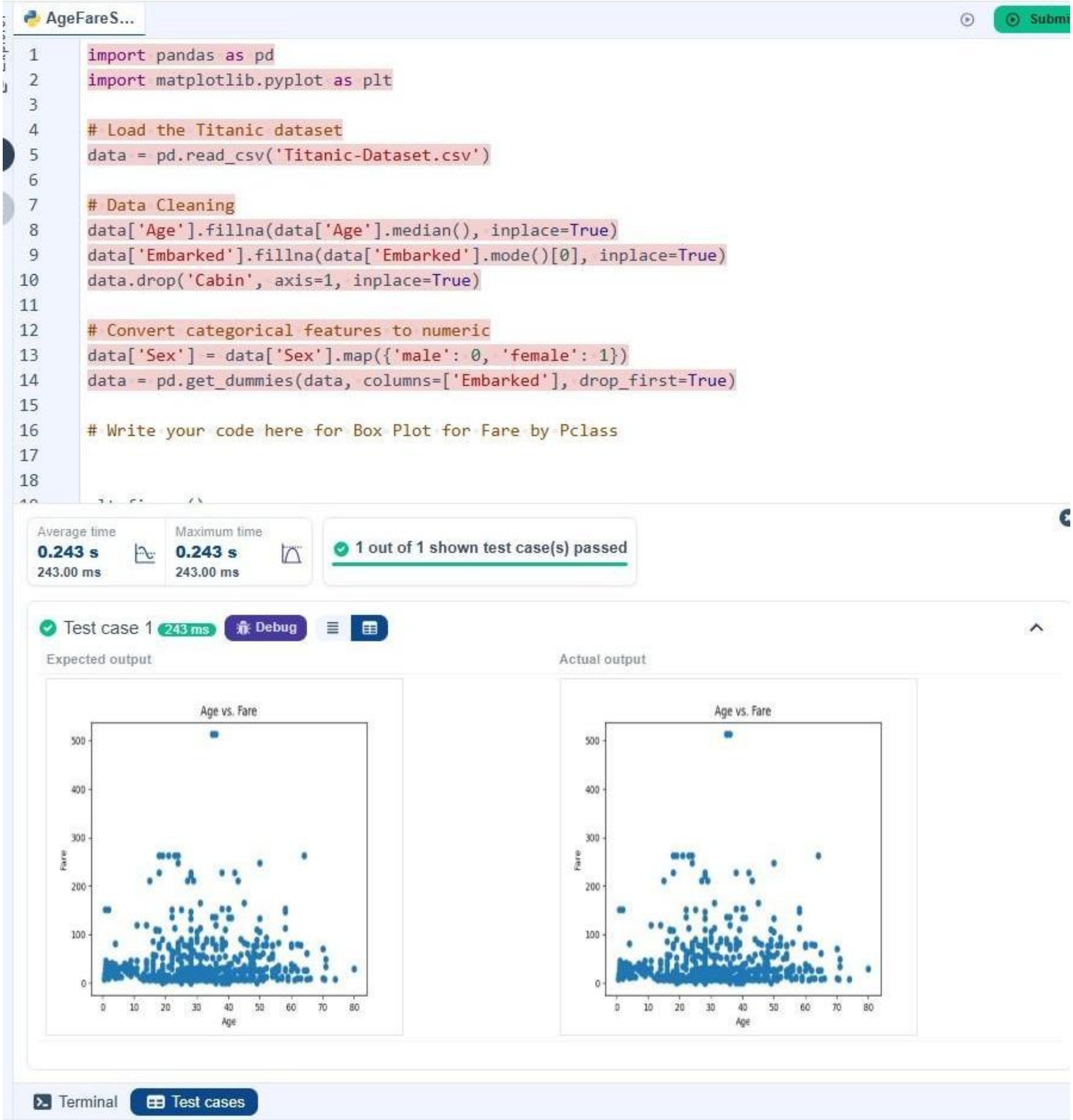
Data Cleaning

```
data['Age'].fillna(data['Age'].median(),

inplace=True)

data['Embarked'].fillna(data['Embarked'

].mode()[0], inplace=True) data.drop('Cabin',

axis=1, inplace=True)
```

```
# Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':

0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],

drop_first=True)
```

```
# Write your code here for Box Plot for Fare by Pclass
```

```
plt.figure() plt.scatter(data['Age'],

data['Fare']) plt.title('Age vs. Fare')

plt.xlabel('Age') plt.ylabel('Fare') plt.show()
```



5.2.11.SCATTER PLOT FOR AGE VS. FARE BY SURVIVED

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. **Use the Age column for the x-axis and the Fare column for the y-axis.**
2. **Color the points based on the Survived column: Red for passengers who did not survive (Survived = 0). Blue for passengers who survived (Survived = 1).**
3. **Set the title of the plot to "Age vs. Fare by Survival".**

4. **Label the x-axis as 'Age' and the y-axis as 'Fare'.**

The Titanic dataset contains columns as shown below,

PassengerId	**Survived Pclass Name Sex Age SibSp Parch Ticket Fare Cabin Embarked**

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Code:

```
import pandas as pd import matplotlib.pyplot

as plt
```

Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')

#

Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'

].mode()[0], inplace=True) data.drop('Cabin',

axis=1, inplace=True)

Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male':

0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'],

drop_first=True)

Write your code here for Scatter Plot for Age vs. Fare by Survived

plt.figure() colors = {0: 'red', 1: 'blue'} plt.scatter(data['Age'], data['Fare'],

c=data['Survived'].apply(lambda x: colors[x])) plt.title('Age vs. Fare by

Survival') plt.xlabel('Age') plt.ylabel('Fare') plt.show()

Explorer

AgeFareS...

Subm

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Scatter Plot for Age vs. Fare by Survived

plt.figure()
```

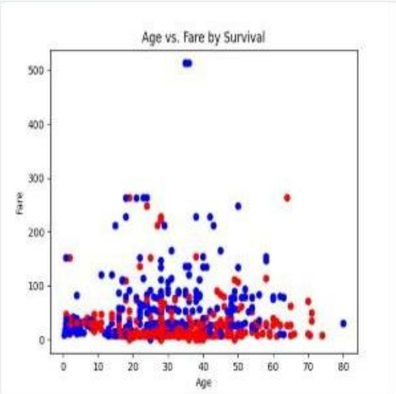
Average time **0.274 s** 274.00 ms

Maximum time **0.274 s** 274.00 ms

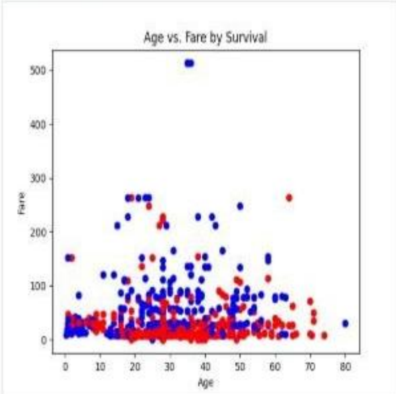
1 out of 1 shown test case(s) passed

Test case 1 274 ms **Debug**

Expected output



Actual output



Terminal

Test cases